

Lesson 09 — Sensors

ROBOTICS BASICS

BEGINNER FRIENDLY

Hey! Welcome to **Lesson 09** — today we're diving into one of the most important topics in robotics: **Sensors**. Without sensors, a robot is basically just a fancy paperweight — it can't see, hear, or feel anything around it. Sensors are how robots interact with the real world.

In this lesson, we'll cover:

- What sensors are and why robots need them
- IR sensors — obstacle detection & line following
- PIR sensors — motion detection using heat
- Ultrasonic sensors — measuring distance with sound
- LDR — light sensing
- Calibration & a handy comparison table

Grab your notebook and let's get into it! 

What is a Sensor?

A **sensor** is a device that **detects changes** in the physical environment — like light, heat, distance, or motion — and converts that into an electrical signal the robot's microcontroller (like an ESP32 or Arduino) can understand and act on.

Why Robots Need Sensors

- To **perceive** their surroundings
- To **avoid obstacles** and not crash
- To **follow lines** or paths
- To **detect motion** (people, objects)
- To **measure distance** accurately
- To **respond to light** conditions
- To make **smart decisions** in real time

Simple Analogy

Think of sensors as a robot's **senses** — just like your eyes, ears, and skin help you understand the world, sensors give a robot its "perception" of the environment.

📋 ★ **Key Point:** No sensors = robot is "blind." Sensors make robots **autonomous!**

Physical Environment

Sensor Detects Change

Microcontroller Acts

IR Sensor — Infrared Sensor

<https://www.youtube.com/watch?v=OMZacCLRt9A>

ACTIVE SENSOR

OBSTACLE DETECTION

An **IR (Infrared) sensor** uses infrared light to detect objects. It has two main parts — an **IR transmitter** (LED that emits IR light) and an **IR receiver** (photodiode that detects reflected IR light).

♦ How It Works — Reflection Principle

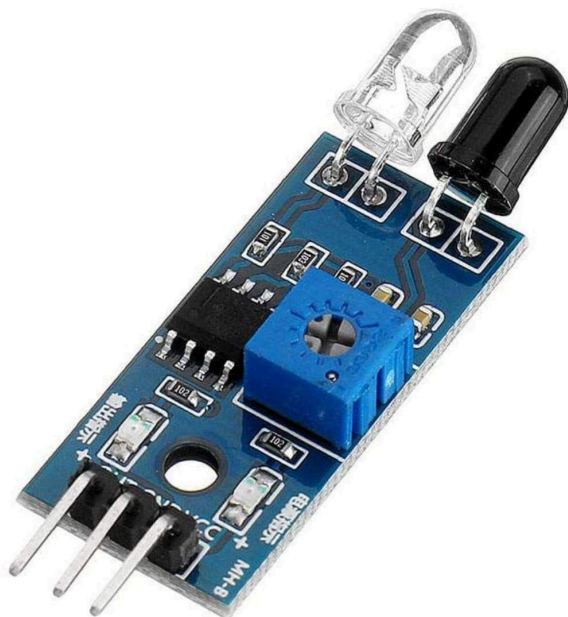
The IR transmitter sends out invisible infrared light. When this light hits an object, it **reflects back**. The receiver detects this reflection. **No reflection** = no object. **Reflection detected** = obstacle present!

♦ Obstacle Detection Mode

Place sensor facing forward on robot. If something is in front → IR reflects back → receiver gets signal → robot stops or turns. Simple and effective! Works well up to ~20–30 cm.

♦ Line Following Mode

Point sensor **downwards** at the ground. **White surface** = reflects IR = HIGH signal. **Black line** = absorbs IR = LOW signal. Robot follows the line by reading these changes continuously.



★ **Active sensor** — it emits its own IR light and doesn't depend on ambient light. Very popular in beginner robotics projects!

PIR SENSOR

PASSIVE SENSOR

MOTION DETECTION

A **PIR (Passive Infrared) sensor** detects **motion** by sensing changes in infrared radiation (heat) emitted by objects — especially warm bodies like humans and animals. It's called "passive" because it **doesn't emit** anything itself — it only **receives**.

Working Principle

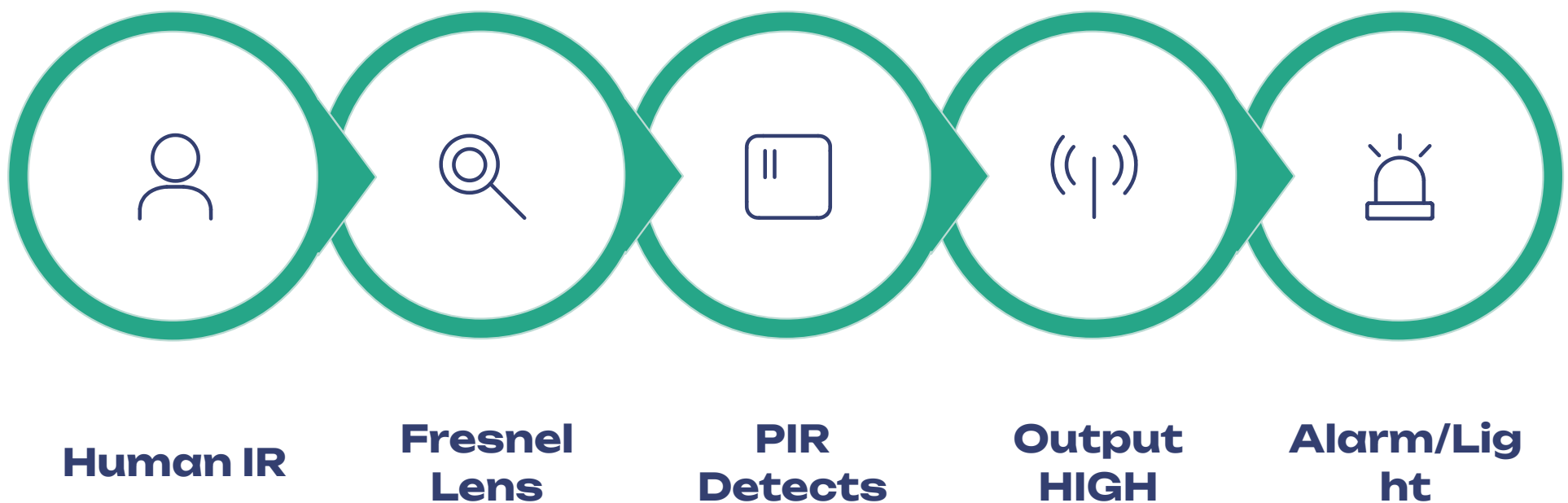
- All objects above absolute zero emit **thermal radiation**
- Humans emit IR around **9–10 μm wavelength**
- PIR has two sensing elements — detects **change** between them
- When a warm body moves → **differential change** detected
- Output goes **HIGH** for a set time

📌 ★ PIR only detects **movement**, not stationary objects!

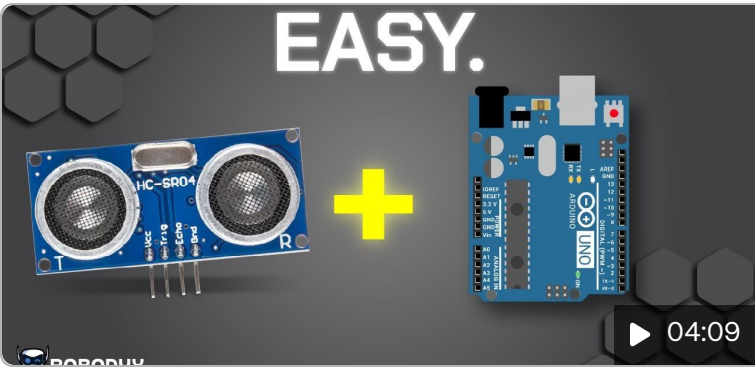
Applications

- Security alarm systems
- Automatic lights (on when you walk in)
- Intruder detection robots
- Smart home automation
- People-counting systems

Common module: **HC-SR501** — easy to use with Arduino/ESP32!



Ultrasonic Sensor — HC-SR04



YouTube

How To Use Ultrasonic Sensors with Arduino! + Project Idea!

Arduino Starter Course & Community 📌 <https://www.skool.com/robonyx/about>
A quick guide on how ultrasonic sensors work, how to use them with Arduino & a...

DISTANCE MEASUREMENT

~40 KHZ SOUND WAVES

The **HC-SR04** is the most popular ultrasonic sensor in student robotics. It measures distance by sending out an ultrasonic pulse (~40 kHz — beyond human hearing) and timing how long the **echo** takes to return.

● TRIG Pin

Send a **10 μ s pulse** from microcontroller to trigger the sensor to emit ultrasonic waves.

● Ultrasonic Wave

Sensor emits **40 kHz sound burst** that travels forward until it hits an object.

● Echo Return

Sound bounces off object and returns. Sensor measures **time of flight**.

● Distance Formula

Distance = (Time $\times$ Speed of Sound) $\div$ 2
Speed of sound $\approx$ 343 m/s

Trigger Pulse

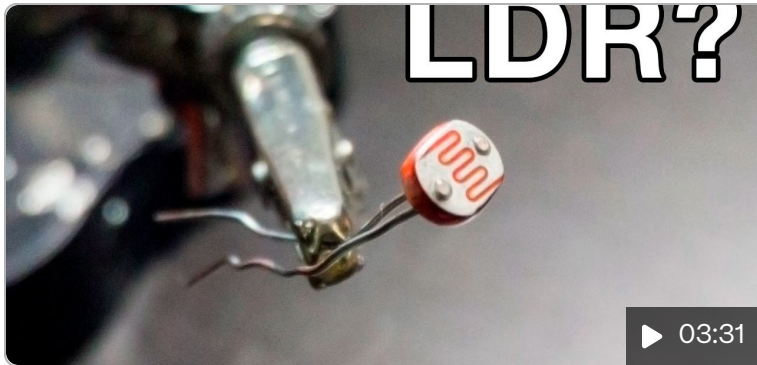
Emit 40kHz Wave

Echo Returns

Measure Duration

★ Range: **2 cm to 400 cm**. Accuracy: ~3mm. Perfect for obstacle-avoidance robots! The " $\div 2$ " in the formula is because sound travels **to the object AND back** — so we only want half the total distance.

LDR — Light Dependent Resistor



YouTube

What is an LDR? (Light Dependant Resistor) – Electronics Basics 26

Let's take a look at LDRs, how they work, and their uses. If you would like to support me to make Simply Electronics even better, you can become a Patron a...

03:31

PASSIVE SENSOR

LIGHT SENSING

An **LDR (Light Dependent Resistor)**, also called a **photoresistor**, is a component whose **resistance changes** based on the intensity of light falling on it. Simple, cheap, and super useful!

How Resistance Changes

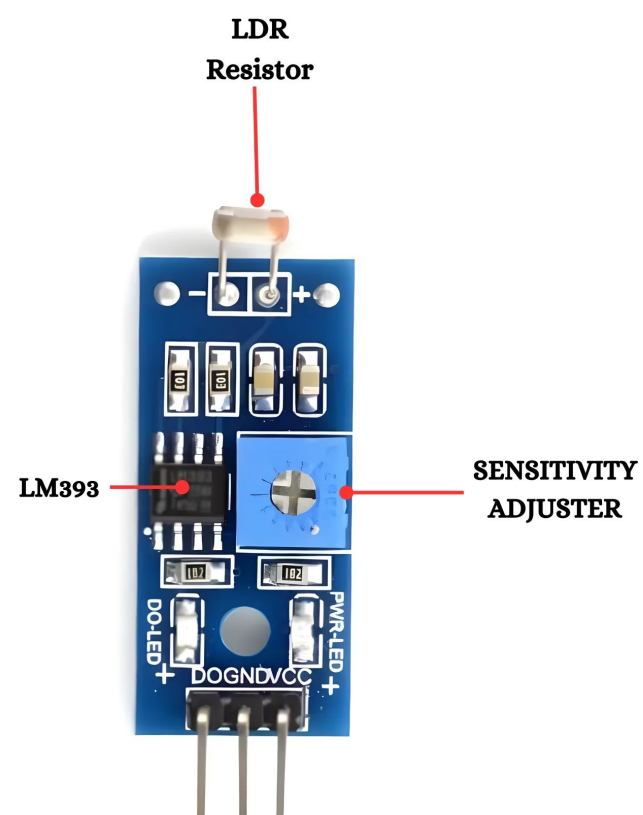
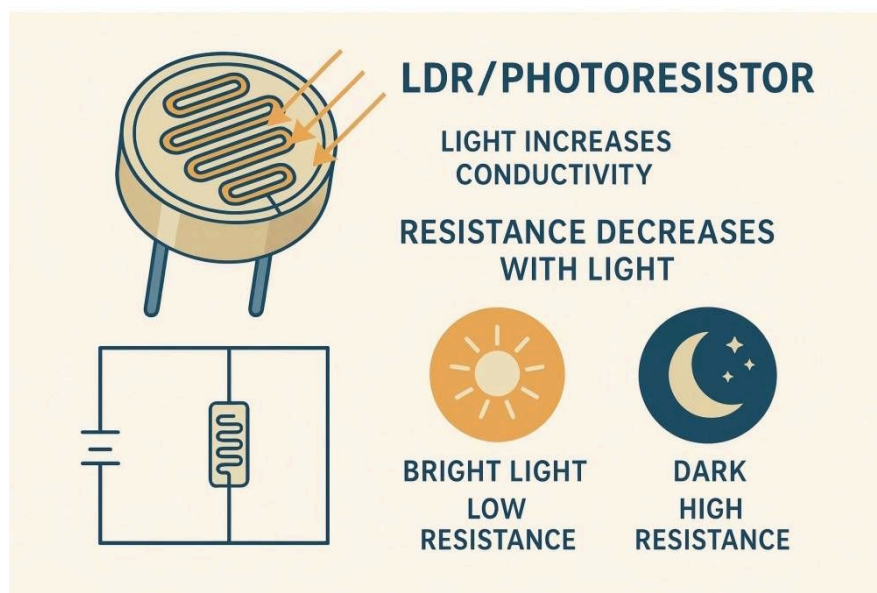
- **Bright light** → Resistance **decreases** (~500 Ω)
- **Darkness** → Resistance **increases** (~1 M Ω)
- More light = more current flows
- Less light = very little current

Used in a **voltage divider circuit** with a fixed resistor to get a readable analog voltage (0–5V) for the microcontroller.

Applications

- Automatic street lights (on at dusk)
- Light-following robots (solar trackers)
- Camera exposure control
- Night/day detection circuits
- Screen brightness auto-adjust

☐ ★ LDRs are **slow to respond** — fine for ambient light, not for fast changes!



Sensor Reading & Calibration

IMPORTANT CONCEPT

PRACTICAL TIP

Raw sensor readings are **never perfect** — they vary based on environment, component tolerances, and noise.

Calibration means testing your sensor in your specific setup and finding the right **threshold values** to make reliable decisions.

1 Read the Raw Values

Open Serial Monitor and watch sensor output in different conditions (bright/dark, near/far, with/without object). Note the range of values you see.

2 Find Your Threshold

Pick a value in the **middle** of your observed range. Example: IR sensor reads 800 in open space and 200 near a wall → threshold ≈ **500**. Above 500 = no obstacle. Below 500 = obstacle!

3 Test & Adjust

Run your robot and observe. If it's too sensitive → adjust threshold. Test in **real conditions** — different lighting, surfaces, distances. Calibration is not a one-time thing!



Why Calibrate? Two identical sensors can give different readings. Your room's lighting is different from the lab. Calibration makes your code work in **your** environment!

Comparison Table

Here's a quick reference table comparing all four sensors we covered. Bookmark this! 

Sensor	Detects	Principle	Example Application
IR Sensor	Objects, lines	IR light reflection	Obstacle avoidance, line follower robot
PIR Sensor	Motion, heat	Thermal radiation change	Security alarm, auto lights, intruder detection
Ultrasonic (HC-SR04)	Distance	Sound echo (40 kHz)	Obstacle mapping, parking sensors, distance measurement
LDR	Light intensity	Resistance changes with light	Auto street lights, light-following robot, day/night detector

Active Sensors IR, Ultrasonic — emit their own signal	Passive Sensors PIR, LDR — only receive, don't emit
Analog Output LDR, some IR — continuous voltage	Digital Output PIR, HC-SR04 — HIGH or LOW signal

Quick Recap & Key Takeaways

Let's wrap up Lesson 09! Here are the most important things to remember before you move on to Lesson 10:

1

Sensors = Robot's Senses

Without sensors, robots are blind. Sensors convert physical world data into electrical signals the microcontroller can read.

2

IR — Reflect & Detect

Transmitter + Receiver. Reflects IR off objects. Used for obstacle detection AND line following. Active sensor.

3

PIR — Detect Heat Motion

Passive sensor. Detects changes in thermal radiation. Only works with *moving* warm objects. Great for security!

4

Ultrasonic — Echo Distance

HC-SR04 sends 40 kHz pulse, times the echo. Distance = $(\text{time} \times 343) \div 2$. Range: 2–400 cm.

5

LDR — Light = Low Resistance

Bright = low resistance. Dark = high resistance. Use voltage divider. Great for light-following robots.

6

Always Calibrate!

Read raw values → find threshold → test in real conditions. Every sensor and environment is different.



Lesson 09 Complete! Next up: Lesson 10 — Motors & Motor Drivers. You're doing great, keep going! 